Assume that the population is tripled in t_1 years.

That is, $x = 3x_0$, when $t = t_1$.

Thus, $t_1 = 50\left(\frac{\log 3}{\log 2}\right)$. Therefore, the population is tripled in $50\left(\frac{\log 3}{\log 2}\right)$ years. ■

10.8.2. Radioactive decay

The nucleus of an atom consists of combinations of protons and neutrons. Many of these combinations of protons and neutrons are unstable, that is the atoms decay or transmute into the atoms of another substance. Such nuclei are said to be **radioactive**.

It is assumed that the rate $\frac{d\mathrm{A}}{dt}$ at which the nuclei of a substance decays is proportional to the amount $\mathrm{A}(t)$ of the substance remaining at time t.

Thus, the required differential equation is $\frac{d\mathrm{A}}{dt} \propto \mathrm{A}$ or $\frac{d\mathrm{A}}{dt} = k\mathrm{A}$ …(2), where k is the constant of proportionality. Here $k < 0$, since decay occurs.

Remarks

From equations (1) and (2), we see that the differential equations are the same, but the difference is only in the interpretations of the symbols and the constants of proportionality. For growth as we expect in (1), $k > 0$ and in the case of (2) for decay, $k < 0$.

A single differential equation can serve as a mathematical model for many different phenomena.

Example 10.28

A radioactive isotope has an initial mass 200mg, which two years later is 50mg. Find the expression for the amount of the isotope remaining at any time. What is its half-life? (half-life means the time taken for the radioactivity of a specified isotope to fall to half its original value).

Solution

Let A be the mass of the isotope remaining after t years, and let $-k$ be the constant of proportionality, where $k > 0$. Then the rate of decomposition is modeled by $\frac{d\mathrm{A}}{dt} = -k\mathrm{A}$, where the minus sign indicates that the mass is decreasing. It is a separable equation. Separating the variables, we get $\frac{d\mathrm{A}}{\mathrm{A}} = -kdt$

Integrating on both sides, we get $\log|\mathrm{A}| = -kt + \log|C|$ or $A = Ce^{-kt}$.

Given that the initial mass is 200mg. That is, $\mathrm{A} = 200$ when $t = 0$ and thus, $C = 200$.

Thus, we get $\mathrm{A} = 200e^{-kt}$.

Also, $\mathrm{A} = 150$ when $t = 2$ and therefore, $k = \frac{1}{2}\log\left(\frac{4}{3}\right)$.

Hence, $\mathrm{A}(t) = 200e^{-\frac{t}{2}\log\left(\frac{4}{3}\right)}$ is the mass of isotope remaining after t years.

Chapter 10 Differential Equation.indd 171 26-02-2020 17:26:06